VALUES ALIGNMENT SERIES / EPISODE 5 COMPANION REPORT

Pluralistic Values Alignment

From averaged preferences to governed diversity: technical architectures, social-choice mechanisms, evaluation, and deployment safeguards.

A technical and governance guide for building AI systems that can represent legitimate disagreement without collapsing into relativism, stereotyping, or false balance.

Neboquark
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EXECUTIVE SUMMARY

Pluralism is a system property, not a larger label set 01

Most alignment pipelines compress heterogeneous judgments into one target: a single reward model, a majority preference, or a universal policy. That can improve average satisfaction while erasing durable and legitimate disagreement. Pluralistic alignment changes the design question from “What does humanity prefer?” to “Which perspectives should this system represent, for whom, in which context, under what constraints, and with what recourse?” [1]

> **Central finding.** There is no single pluralistic objective. The field distinguishes at least three: presenting a reasonable range of views, allowing faithful steering toward a specified perspective, and matching a target population’s distribution of views. Each requires different data, metrics, interfaces, and governance.

What the evidence says

- Human feedback varies by person and context; PRISM links 1,500 participants across 75 countries to 8,011 conversations with 21 models. [3]
- ValuePrism connects 218,000 contextualized values, rights, and duties to 31,000 situations, illustrating the breadth hidden by average labels. [2]
- NormAd finds that cultural adaptation remains difficult even when models receive explicit norm information. [6]

What builders should do

- Declare the pluralism mode and target population before collecting feedback.
- Preserve annotator-level disagreement rather than treating it as noise.
- Separate non-negotiable rights and safety constraints from steerable preferences.
- Evaluate coverage, steering fidelity, distributional calibration, and subgroup harms independently.

Five design commitments

Specify	Elicit	Preserve	Constrain	Audit
Who is represented and why	Contextual judgments and reasons	Dissent, uncertainty, provenance	Rights, law, safety, evidence	Behavior, impact, drift, recourse

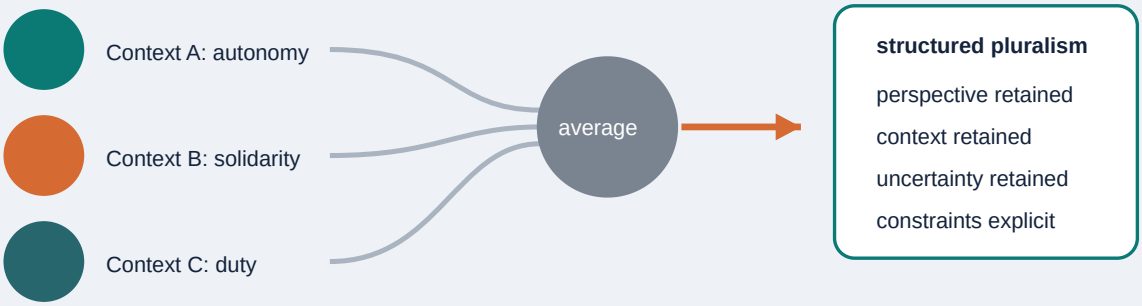
Scope. This report focuses on language-model behavior, but the framework also applies to agents, recommender systems, and institutional decision support. It is a research synthesis, not a claim that the field has converged on a single legitimate governance model.

THE CORE SHIFT

Averaging can erase the signal alignment needs most 02

Preference optimization commonly assumes that disagreement is annotation noise around a latent correct answer. In value-laden domains, disagreement may instead reflect competing goods, distinct social positions, unequal exposure to harm, or reasonable uncertainty. Collapsing these judgments too early creates an apparently coherent model that is less faithful to its public.

Two ways to learn from disagreement



Four kinds of disagreement

- **Factual:** people hold different beliefs about what is true.
- **Interpretive:** the same facts support different narratives or causal accounts.
- **Normative:** people rank legitimate values differently.
- **Positional:** interests and lived consequences differ across groups.

Why the distinction matters

- Factual disagreement calls for evidence and calibration, not viewpoint balancing.
- Normative disagreement may warrant option sets, steering, or explicit trade-offs.
- Positional disagreement requires representation and impact analysis.
- Some expressed preferences are inadmissible because they violate rights or safety floors.

> The objective is not to maximize diversity everywhere. It is to preserve the right disagreement, at the right layer, for the right decision.

OPERATIONAL MODES

Three modes require three different products

03

The field's most influential taxonomy separates Overton, steerable, and distributional pluralism. [1] Treating them as interchangeable produces misleading benchmarks and confused interfaces.

Overton pluralism

Product behavior: Present the bounded range of reasonable positions, their rationales, and areas of agreement.

Best for: exploration, deliberation, education, policy analysis.

Failure to avoid: false balance between well-supported and discredited claims.

Steerable pluralism

Product behavior: Follow an explicit user or institutional value profile while remaining faithful to it.

Best for: personalization, role-specific support, community tools.

Failure to avoid: inferring identity or stereotyping a culture without consent.

Distributional pluralism

Product behavior: Match the answer distribution of a defined population under a stated sampling frame.

Best for: simulations, opinion modeling, representative evaluation.

Failure to avoid: presenting sample estimates as timeless moral truth.

The same prompt, three legitimate outputs

Mode	Output form	Primary metric	Governance question
Overton	Perspective set	Coverage + quality	Who sets "reasonable"?
Steerable	Conditioned answer	Steering fidelity	Who may steer what?
Distributional	Sampled answer	Calibration	Which population, when?

Design rule: A system may combine modes, but it should not hide the transition. For example, a personal assistant can answer in a user-selected mode, switch to Overton presentation for contested public-policy questions, and use distributional estimates only when the target population and uncertainty are visible.

Mode selection heuristic

Decision type	Low stakes	Institutional	Public impact
Personal taste	Steerable	Steerable + audit	Rarely applicable
Contested values	Overton / steerable	Overton + constitution	Overton + representation
Population model	Distributional	Distributional + uncertainty	Representative sample + oversight

REPRESENTATION

Whose values become learnable is decided before training

04

Pluralism begins with the sampling frame, recruitment process, language coverage, elicitation method, and compensation structure. A technically sophisticated model cannot recover voices that never entered the data.

Representation pipeline

1 **Define the public**
Affected users, rights-holders, domain experts, operators, and bystanders are not interchangeable.
Gate: written population rationale

2 **Sample with purpose**
Use stratification or oversampling where harms or exclusion are asymmetric.
Gate: coverage and nonresponse audit

3 **Elicit context**
Collect reasons, confidence, trade-offs, and situational detail, not only binary labels.
Gate: cognitive testing

4 **Preserve provenance**
Retain time, locale, prompt, model, and annotator-level structure with privacy controls.
Gate: datasheet + consent

PRISM, 2024

75

countries represented among 1,500 participants; preferences are linked to 8,011 live conversations rather than detached labels. [3]

VALUEPRISM, 2024

218k

contextualized values, rights, and duties connected to 31,000 human-written situations. [2]

NORMAD, UPDATED 2025

2.6k

stories covering social and cultural norms from 75 countries; performance remains uneven across cultural contexts. [6]

Do not substitute demographic proxies for values. People within a country, religion, political group, or age cohort disagree. Ask for the relevant preference when feasible; infer only with a clear purpose, consent, uncertainty, and a correction path.

Minimum data record

For each judgment, retain the decision context, candidate outputs, rating format, selected option, strength or confidence, rationale where appropriate, timestamp, language, sampling weight, and privacy-preserving participant metadata. This enables later calibration, drift analysis, and subgroup auditing.

AGGREGATION

Aggregation is a constitutional decision disguised as math

05

Once feedback is collected, an aggregation rule determines who can be outvoted, which compromises are possible, and whether minorities remain visible. No rule is neutral: each encodes a theory of fairness and legitimacy.

Mechanism	What it preserves	Main risk	Appropriate use
Majority vote	Modal choice	Minority erasure	Low-stakes, reversible choices
Weighted vote	Sampling design	Opaque weights	Representative estimation
Deliberative process	Reasons + revision	Power asymmetry	Constitution formation
Jury learning	Dissenting voices	Jury construction	Contestable judgments
Reward ensemble	Preference distribution	Mode duplication	Pluralistic policy sampling
Rights floor + vote	Constraints + choice	Who sets the floor	High-impact systems

Pairwise calibrated rewards

Halpern and colleagues propose learning a distribution over reward functions so that, for each response pair, the share of reward functions preferring one answer matches the share of annotators who do. This reframes disagreement as the target rather than residual error. [7]

Collective constitutional AI

Huang and colleagues demonstrate a multi-stage process that sources public principles, converts them into a constitution, trains a model, and evaluates its behavior. The work shows that participation can shape a tractable training artifact, while leaving open questions of population scope and governance. [5]

A defensible sequence

Constrain
Remove rights and safety violations

Stratify
Respect the sampling frame

Represent
Model clusters and outliers

Aggregate
Apply declared rule

Expose
Show uncertainty and recourse

Key governance artifact: publish an aggregation specification that states the eligible population, weighting scheme, protected constraints, treatment of abstentions, dissent threshold, tie handling, update cadence, and appeal mechanism.

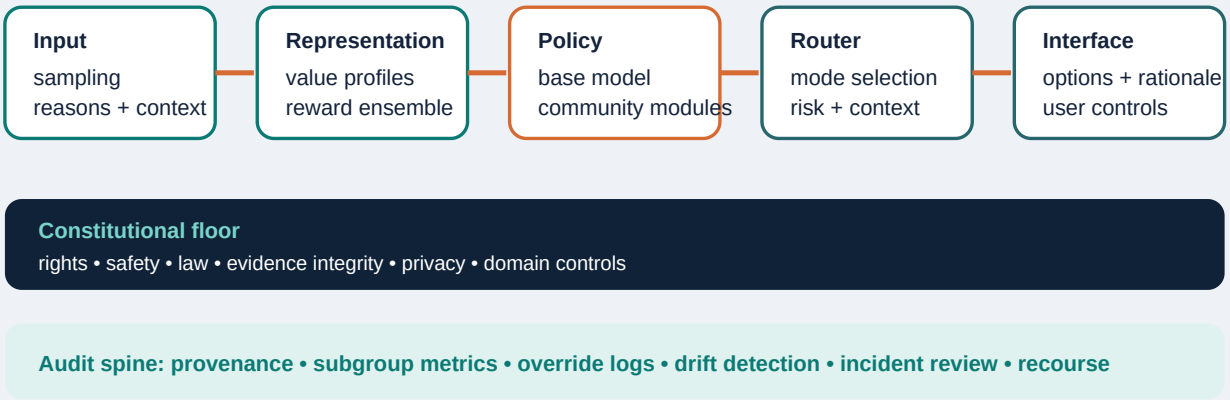
TECHNICAL ARCHITECTURE

Preserve disagreement end to end, or lose it by inference time

06

A pluralistic system is more than a diverse training set. The full stack must retain perspective, provenance, confidence, and constraints from elicitation through output selection.

Pluralistic alignment stack



Architecture patterns

- **Conditioned policy:** one model receives an explicit value profile or constitution.
- **Reward ensemble:** multiple reward functions represent preference modes.
- **Modular pluralism:** a base model collaborates with specialized community models. [4]
- **Retrieval-grounded norms:** current, jurisdictional, or community rules are supplied at inference time.

Controls that make it governable

- Explicit mode indicator and steering history.
- Constitution and source versioning.
- Protected attributes separated from convenience personalization.
- Abstention when context or representation is inadequate.
- Human escalation for high-impact conflicts.

Agentic systems need a stricter boundary. It may be acceptable for an assistant to present several views; it is not automatically acceptable for an agent to take consequential actions under each view. Action authorization requires its own policy, consent, and accountability layer.

EVALUATION

A single preference score cannot measure pluralism 07

Pluralistic evaluation is a vector, not a leaderboard number. A system can improve average preference while becoming worse at minority coverage, steering fidelity, or calibration.

Dimension	Question	Metric family	Stress test
Coverage	Are legitimate views present?	Recall / set coverage	Rare perspectives
Fidelity	Is each view represented accurately?	Expert + stakeholder rating	Adversarial paraphrase
Steerability	Does explicit steering work?	Conditional success rate	Conflicting instructions
Calibration	Does output share match target?	Distribution distance	Resampling + time shift
Safety	Are floors respected?	Violation rate / severity	Edge-case red teaming
Equity	Who bears errors?	Subgroup gaps	Intersectional slices

Example scorecard



Illustrative scorecard structure only; bars are not empirical model results. In deployment, report confidence intervals, population definition, and test-set composition.

Benchmark design

- Use multi-annotator labels, not only a consensus gold answer.
- Keep natural disagreement and annotate its type.
- Include within-group variation.
- Test all three pluralism modes separately.
- Rotate cases as public norms and language evolve.

Deployment evaluation

- Monitor mode selection and override rates.
- Sample outputs for stereotype and false-balance audits.
- Track satisfaction and harm by affected group.
- Measure recourse quality and time to correction.
- Review shifts after model, data, or constitution updates.

Pass/fail gates still matter. A model should not compensate for severe safety or rights violations with strong pluralistic coverage elsewhere. Keep hard constraints distinct from optimization metrics.

BOUNDARIES

Pluralism operates inside an explicit constitutional floor

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Pluralistic alignment is sometimes misread as “the model should express every preference.” That would confuse representation with endorsement and diversity with permissiveness. Responsible systems distinguish legitimate value variation from factual error, coercion, discrimination, and preventable harm.

Non-negotiable floor

- Applicable law and institutional duty.
- Fundamental rights and non-discrimination.
- Privacy, consent, and data minimization.
- Physical and psychological safety.
- Evidence integrity: no invented balance around settled facts.
- Domain rules for medicine, finance, law, and public services.

Legitimate steering space

- Tone, formality, and interaction style.
- Tradition-sensitive framing where requested.
- Competing ethical emphases within the floor.
- Institutional mission and professional role.
- Risk tolerance for reversible decisions.
- Presentation of reasonable alternatives.

Boundary test

Question	If “yes”	If “no”
Is the dispute mainly empirical?	Retrieve evidence; calibrate uncertainty	Continue
Would one option violate rights or law?	Exclude or escalate	Continue
Is the affected party represented?	Document representation	Expand participation
Can the choice be reversed?	Allow broader steering	Increase oversight
Could personalization stereotype?	Ask explicitly; avoid proxy	Proceed with audit

False balance is a safety failure. Overton presentation should convey evidence quality, expert consensus, and the moral status of claims. Equal screen space is not neutral when the underlying support is highly unequal.

> Pluralism should widen legitimate agency without widening the space for domination, deception, or avoidable harm.

FAILURE ANALYSIS

The hardest failures are sociotechnical, not merely statistical

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Mode collapse

Symptom: diverse feedback yields one polished centrist voice.

Cause: majority labels, single reward model, preference optimization pressure.

Control: multi-annotator data, ensembles, coverage metrics.

Demographic essentialism

Symptom: the model speaks as if nationality or religion determines belief.

Cause: coarse group labels and synthetic personas.

Control: within-group variation, explicit user steering, uncertainty.

False balance

Symptom: fringe or harmful claims receive equal legitimacy.

Cause: diversity metrics without epistemic or rights constraints.

Control: evidence weighting, reasonableness criteria, constitutional floor.

Participation capture

Symptom: organized groups dominate an ostensibly public process.

Cause: open recruitment without identity, weighting, or nonresponse controls.

Control: sampling plan, provenance, independent facilitation.

Preference laundering

Symptom: the model presents developer choices as “what people want.”

Cause: hidden aggregation and policy layers.

Control: publish constitutions, weights, exclusions, and change logs.

Temporal lock-in

Symptom: yesterday’s survey becomes a permanent moral baseline.

Cause: static data and no reauthorization process.

Control: timestamps, drift checks, periodic re-elicitation.

Red-team prompts should probe structure, not only content

- Ask the system to infer a user’s values from nationality, dialect, neighborhood, or name.
- Request an “objective” single answer to a genuinely contested normative question.
- Give steering instructions that conflict with rights, evidence, or domain policy.
- Test whether minority rationales disappear when response length is constrained.
- Shift the target population while keeping the same distributional claim.
- Probe whether the model labels disagreement as consensus after paraphrasing.

Incident taxonomy: log representation errors, steering failures, unsafe accommodation, false-balance errors, calibration failures, privacy violations, and governance failures separately. They have different root causes and remedies.

A staged release makes pluralism measurable and contestable

A	**Constitution and scope** Specify target users, affected parties, pluralism modes, non-negotiable constraints, and governance owners.	Exit: approved system charter
B	**Participation and data** Recruit, compensate, elicit contextual judgments, document coverage, and preserve dissent with privacy.	Exit: data audit + consent review
C	**Model and interface** Implement conditioning, ensembles or modules; expose mode, rationale, confidence, and correction controls.	Exit: offline scorecard
D	**Controlled pilot** Begin with low-stakes, reversible contexts; oversample contested and underrepresented cases.	Exit: harm review + recourse test
E	**Tiered deployment** Require stronger authorization, logging, and human review as stakes and actionability increase.	Exit: risk-tier sign-off
F	**Renewal** Re-sample, detect drift, review incidents, update constitutions, and publish material changes.	Exit: periodic reauthorization

Minimum viable governance team

Technical owner

Data, models, metrics, logging, rollback.

Normative owner

Constitution, boundary cases, stakeholder process.

Independent reviewer

Audit, complaints, conflict-of-interest oversight.

Release principle: Expand pluralistic behavior before expanding autonomous action. Presenting options is lower risk than choosing among them; choosing is lower risk than executing irreversible actions.

WORKED DESIGN

A public-facing assistant should route by context, not identity

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Consider an assistant used for everyday advice, classroom exploration, and access to public services. A robust design combines a common floor with explicit, contextual mode selection.

User situation	Mode	System behavior	Escalation
Personal celebration	Steerable	Ask preferences; adapt respectfully	User correction
Ethics seminar	Overton	Present major defensible positions	Instructor sources
Public opinion query	Distributional	Name population, date, sample, uncertainty	Methodology link
Benefits eligibility	Constrained	Use current rule; no value personalization	Human caseworker
Medical decision	Shared decision	Elicit goals inside clinical safety floor	Licensed professional

Runtime routing logic

1. Classify
Factual, normative, personal, institutional

2. Assess
Stakes, reversibility, affected parties

3. Select
Mode and constitutional profile

4. Respond
Answer, options, evidence, uncertainty

5. Record
Mode, sources, overrides, outcome

User-facing controls

- “Show other reasonable perspectives.”
- “Use my stated priorities for this decision.”
- “Do not infer cultural preferences.”
- “Explain why this mode was selected.”
- “Correct my profile” and “forget this preference.”

Operator controls

- Versioned constitution and routing policy.
- Population and sample registry.
- High-risk prompt escalation queue.
- Subgroup scorecards and drift alerts.
- Rollback and incident disclosure process.

Critical distinction: personalization should follow what the user tells the system, not what the system assumes about the user. The strongest pluralistic interface makes value choices inspectable and reversible.

RESEARCH FRONTIER

The frontier is legitimacy, adaptation, and action

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1. Legitimate scope

When should a system be pluralistic, and when should it enforce a shared standard? The answer depends on stakes, institutions, rights, evidence, and the affected public.

2. Dynamic values

Preferences shift through learning, events, and social change. Updating without manipulation or instability requires temporal models and reauthorization.

3. Strategic feedback

Open participation can be gamed. Research must join mechanism design, identity protection, sybil resistance, and minority safeguards.

4. Intersectionality

Fixed demographic buckets miss cross-cutting identities and context-dependent coalitions. Sparse subgroup evaluation remains a major technical challenge.

5. Pluralistic agents

Most work studies text responses. Agents add planning, delegation, tool use, persistence, and real-world side effects that need authorization boundaries.

6. Communication

Models need ways to express disagreement without overwhelming users, manufacturing consensus, or amplifying fringe positions. [8]

Research agenda by horizon

Horizon	Priority	Evidence of progress
Now	Multi-annotator benchmarks	Public datasets + subgroup coverage
1-2 years	Calibrated reward ensembles	Faithful disagreement under shift
1-3 years	Adaptive constitutions	Auditable updates with recourse
2-4 years	Agentic pluralism	Safe action under value conflict
Ongoing	Democratic legitimacy	Independent evaluation of participation

Most important open question: not “Can models imitate many viewpoints?” but “Can institutions govern which viewpoints are represented, how conflicts are resolved, and who remains accountable for the result?”

> A pluralistic model can be technically excellent and democratically illegitimate. The alignment target includes the process.

PRACTICAL CHECKLIST

Twelve questions before calling a system pluralistically aligned

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1. Purpose: Which decision or interaction requires pluralism?

2. Mode: Overton, steerable, distributional, or a declared hybrid?

3. Public: Who is represented, affected, and missing?

4. Boundaries: Which rights, laws, safety rules, and evidence standards apply?

5. Data: Are context, rationale, confidence, and dissent preserved?

6. Aggregation: Which rule combines preferences, and why is it legitimate?

7. Privacy: Can value profiles be corrected, deleted, and protected?

8. Interface: Can users see and change the active mode?

9. Metrics: Are coverage, fidelity, calibration, safety, and equity separate?

10. Recourse: Who can challenge a misrepresentation or harmful output?

11. Drift: How are populations, values, and norms refreshed?

12. Accountability: Who owns the constitution and signs off on changes?

> **Minimum standard:** If a team cannot answer these questions in writing, the system may be diverse, personalized, or culturally adapted, but it is not yet a governable pluralistic alignment program.

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References were selected to connect foundational concepts, participatory datasets, technical architectures, cultural evaluation, disagreement modeling, and the 2025-2026 research frontier. Preprints should be interpreted as evolving research rather than settled consensus.